Qiao Liu

Contact Information

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New Haven, CT, 06511

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Research Interest

Generative AI; Causal Inference; Gene Regulation; Multiomics Integration; Single Cell Genomics; Genomic Foundation Models; Pharmacogenomics;

Professional Experience

Assistant Professor — Sep 2025 - Present
Department of Biostatistics — New Haven, CT
Yale University

Postdoctoral Scholar — Jul 2021 - Aug 2025
Department of Statistics — Stanford, CA
Advisor: Prof. Wing Hung Wong

Education

Stanford University — Aug 2019 - Jul 2021
Visiting Ph.D., Department of Statistics — Stanford, CA
Advisor: Prof. Wing Hung Wong

Tsinghua University — Sep 2016 - Aug 2019
Ph.D. student, Department of Automation — Beijing, China
Advisor: Prof. Rui Jiang

Beihang University — Sep 2012 - Aug 2016
B.E. (with honors), ShenYuan Honors College — Beijing, China

Manuscripts Submitted

(†=co-first author; *=corresponding author)

1. Xuejian Cui, Haochen Wang, Zijing Gao, Rui Jiang*, Wing Hung Wong*, **Qiao Liu***. scMTG reconstructs single-cell temporal dynamics with Markov transition generators[J]. arXiv, 2026. [preprint]

2. **Qiao Liu***. Missingness-aware data imputation via AI-powered Bayesian generative modeling[J]. arXiv, 2026. [preprint]

3. Guyue Luo and **Qiao Liu***. BGM-IV: an AI-powered Bayesian generative modeling approach for instrumental variable analysis[J]. arXiv, 2026. [preprint]

4. **Qiao Liu*** and Wing Hung Wong. A Bayesian generative modeling approach for arbitrary conditional inference[J]. arXiv, 2026. [preprint]

5. Erica Stutz, Giacomo Marino, Daniella Meeker, **Qiao Liu**, Andrew J. Loza. Conditional attribute estimation with autoregressive sequence models[J]. arXiv, 2026. [preprint]

6. **Qiao Liu**†, Wanwen Zeng†, Hongtu Zhu, Lexin Li*, Wing Hung Wong*. Leveraging genomic large language models to enhance causal genotype-brain-clinical pathways in Alzheimer's disease[J]. medRxiv, 2025. [preprint]

Peer-reviewed Publications

1. **Qiao Liu*** and Wing Hung Wong. An AI-powered Bayesian generative modeling approach for causal inference in observational studies[J]. ***Journal of the American Statistical Association***, 2026. `https://doi.org/10.1080/01621459.2026.2654227`

2. Erpai Luo, Lei Wei, Minsheng Hao, Xuegong Zhang*, **Qiao Liu***. A multi-modal diffusion model with dual-cross-attention for multi-omics data generation and translation[J]. ***Nature Communications***, 2026. `https://doi.org/10.1038/s41467-026-71744-x`

3. Weihua Zeng, Chun-Chi Liu, Shuo Li, Yonggang Zhou, Mary L. Stackpole, Ying Xiao, Ran Hu, Caitlin Tang, **Qiao Liu**, . . . , Xianghong Jasmine Zhou. Toward the simultaneous detection of multiple diseases with a highly cost-effective cell-free DNA methylome test [J]. ***Proceedings of the National Academy of Sciences***, 2026. `https://doi.org/10.1073/pnas.2518347123`

4. Xuejian Cui, Qijin Yin, Zijing Gao, Zhen Li, Xiaoyang Chen, Shengquan Chen, **Qiao Liu**, Wanwen Zeng*, Rui Jiang*. CREATE: cell-type-specific cis-regulatory elements identification via discrete embedding[J]. ***Nature Communications***, 2025. `https://doi.org/10.1038/s41467-025-59780-5`

5. Yijiang Liu, Feifan Zhang, Yifei Ge, **Qiao Liu**, Siyu He, Xiaotao Shen. Application of LLMs/Transformer-Based models for metabolite annotation in metabolomics[J]. ***Health and Metabolism***, 2025. `https://doi.org/10.53941/hm.2025.100014`

6. Zhaoyang Zhang, Ziqi Chen, **Qiao Liu**, Jinhan Xie, Hongtu Zhu. Sampling-guided heterogeneous graph neural network with temporal smoothing for scalable longitudinal data imputation[C]. Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (***KDD***), 2025. `https://doi.org/10.1145/3711896.3737116`

7. Wanwen Zeng, Hanmin Guo, **Qiao Liu**, Wing Hung Wong. How to improve polygenic prediction from whole-genome sequencing data by leveraging predicted epigenomic features?[J]. ***Proceedings of the National Academy of Sciences***, 2025. `https://doi.org/10.1073/pnas.2419202122`

8. Wei Zhang, Xianglin Zhang, **Qiao Liu**, Lei Wei, Xu Qiao, Rui Gao, Zhiping Liu, Xiaowo Wang. Deconer: An evaluation toolkit for reference-based deconvolution methods using gene expression data[J]. ***Genomics, Proteomics & Bioinformatics***, 2025. https://doi.org/10.1093/gpbjnl/qzaf009

9. Zijing Gao†, **Qiao Liu**†*, Wanwen Zeng, Wing Hung Wong*, Rui Jiang*. EpiGePT: a Pretrained Transformer model for human epigenomics[J]. ***Genome Biology***, 2024. https://doi.org/10.1186/s13059-024-03449-7

10. **Qiao Liu**, Zhongren Chen, Wing Hung Wong*. An encoding generative modeling approach to dimension reduction and covariate adjustment in causal inference with observational studies [J]. Proceedings of the National Academy of Sciences of the United States of America(***Proceedings of the National Academy of Sciences***), 2024, 121(23), e2322376121. https://doi.org/10.1073/pnas.2322376121

11. Wei Shao, Yuti Liu, Shuang Zhang, Shuang Chen, **Qiao Liu**, Jianyu Zhou, Wanwen Zeng. Inferring gene regulatory network based on scATAC-seq data with gene perturbation[C]. IEEE International Conference on Bioinformatics and Biomedicine (***BIBM***), 2024. https://doi.org/10.1109/bibm62325.2024.10822868

12. Chuanqi Lao, Pengfei Zheng, Hongyang Chen, **Qiao Liu**, Feng An, Zhao Li. DeepAEG: a model for predicting cancer drug response based on data enhancement and edge-collaborative update strategies[J]. ***BMC bioinformatics***, 25 (1), 105, 2024. https://doi.org/10.1186/s12859-024-05723-8

13. Shuo Li, Weihua Zeng, Xiaohui Ni, **Qiao Liu**, Wenyuan Li, Mary L. Stackpole, Yonggang Zhou, Arjan Gower, Kostyantyn Krysan, Preeti Ahuja, David S. Lu, Steven S. Raman, William Hsu, Denise R. Aberle, Clara E. Magyar, Samuel W. French, Steven-Huy B. Han, Edward B. Garon, Vatche G. Agopian, Wing Hung Wong*, Steven M. Dubinett* and Jasmine Zhou*. Comprehensive tissue deconvolution of cell-free DNA by deep learning for disease diagnosis and monitoring[J]. Proceedings of the National Academy of Sciences of the United States of America(***Proceedings of the National Academy of Sciences***), 120 (28) e2305236120, 2023. https://doi.org/10.1073/pnas.2305236120

14. Shuang Zhang, Yuti Liu, Shuang Chen, **Qiao Liu**, Wanwen Zeng. Applications of Transformer-based language models in bioinformatics: A survey[J]. ***Bioinformatics Advances***, 2023. https://doi.org/10.1093/bioadv/vbad001

15. **Qiao Liu**†, Wanwen Zeng†, Wei Zhang, Sicheng Wang, Hongyang Chen, Rui Jiang*, Mu Zhou*, Shaoting Zhang*. Deep generative modeling and clustering of single cell Hi-C data[J]. ***Briefings in Bioinformatics***, 2023, 24(1): bbac494. `https://doi.org/10.1093/bib/bbac494`

16. Qijin Yin, Xusheng Cao, Rui Fan, **Qiao Liu***, Rui Jiang* and Wanwen Zeng*. DeepDrug: A general graph-based deep learning framework for drug-drug interactions and drug-target interactions prediction[J]. ***Quantitative Biology***, 2023, 11(3), 260-274. `https://doi.org/10.15302/J-QB-022-0320`

17. Wanwen Zeng†, **Qiao Liu**†, Qijin Yin†, Rui Jiang*, Wing Hung Wong*. HiChIPdb: a comprehensive database of HiChIP regulatory interactions[J]. ***Nucleic Acids Research***. 2022. `https://doi.org/10.1093/nar/gkac859`

18. Christopher Lance, Malte D. Luecken, Daniel B. Burkhardt, Robrecht Cannoodt, Pia Rautenstrauch, Anna Laddach, Aidyn Ubingazhibov, Zhi-Jie Cao, Kaiwen Deng, Sumeer Khan, **Qiao Liu**, Nikolay Russkikh, Gleb Ryazantsev, Uwe Ohler, NeurIPS 2021 Multimodal data integration competition participants, Angela Oliveira Pisco*, Jonathan Bloom*, Smita Krishnaswamy*, Fabian J. Theis*. Multimodal single cell data integration challenge: results and lessons learned[J]. Proceedings of Machine Learning Research(***PMLR***), 2022. [proc]

19. Zhana Duren, Fengge Chang, Fnu Naqing, Jingxue Xin, **Qiao Liu**, Wing Hung Wong*. Regulatory analysis of single cell multiome gene expression and chromatin accessibility data with scREG[J]. ***Genome Biology***, 2022, 23(1): 1-19. `https://doi.org/10.1186/s13059-022-02682-2`

20. **Qiao Liu**, Kui Hua, Xuegong Zhang, Wing Hung Wong*, Rui Jiang*. DeepCAGE: incorporating transcription factors in genome-wide prediction of chromatin accessibility[J]. ***Genomics, Proteomics & Bioinformatics***, 2022. `https://doi.org/10.1016/j.gpb.2021.08.015`

21. Jinxiang Ou, Yunheng Shen, Feng Wang, **Qiao Liu**, Xuegong Zhang, Hairong Lv*. AggEnhance: Aggregation enhancement by class interior points in federated learning with non-IID data[J]. ACM Transactions on Intelligent Systems and Technology (***TIST***), 2022. `https://doi.org/10.1145/3544495`

22. Qijin Yin, **Qiao Liu**, Zhuoran Fu, Rui Jiang*. scGraph: a graph neural network-based approach to automatically identify cell types[J]. ***Bioinformatics***, 2022. `https://doi.org/10.1093/bioinformatics/btac199`

23. Tianxing Ma, **Qiao Liu**, Haochen Li, Mu Zhou, Rui Jiang, Xuegong Zhang*. DualGCN: a dual graph convolutional network model to predict cancer drug response[J]. ***BMC bioinformatics***, 2022, 23(4): 1-13. `https://doi.org/10.1186/s12859-022-04664-4`

24. **Qiao Liu**, Shengquan Chen, Rui Jiang*, Wing Hung Wong*. Simultaneous deep generative modeling and clustering of single cell genomic data[J]. ***Nature Machine Intelligence***, 2021, 3(6): 536-544. `https://doi.org/10.1038/s42256-021-00333-y`

25. **Qiao Liu**, Jiaze Xu, Rui Jiang*, Wing Hung Wong*. Density estimation with deep generative neural networks [J]. Proceedings of the National Academy of Sciences of the United States of America(***Proceedings of the National Academy of Sciences***), 2021, 118(15), e2101344118. `https://doi.org/10.1073/pnas.2101344118`

26. Feng Wang, Guoyizhe Wei, **Qiao Liu**, Jinxiang Ou, Xian Wei, Hairong Lv*. Boost neural networks by checkpoints [C]. Conference on Neural Information Processing Systems (***NeurIPS***), 2021, 33. `https://doi.org/10.48550/arXiv.2110.00959`

27. Shengquan Chen, **Qiao Liu**, Xuejian Cui, Rui Jiang*. OpenAnnotate: a web server to annotate the chromatin accessibility of genomic regions[J]. ***Nucleic Acids Research***, 2021, 49(W1): W483-W490. `https://doi.org/10.1093/nar/gkab337`

28. **Qiao Liu**, Zhiqiang Hu, Rui Jiang*, Mu Zhou*. Cancer drug response prediction via a hybrid graph convolutional network[J]. ***Bioinformatics***, 2020. 36(Supplement_2): i911-i918. Also in *Proceedings of the 19th European Conference on Computational Biology* (***ECCB***), 2020. `https://doi.org/10.1093/bioinformatics/btaa822`

29. Kexin Ding, **Qiao Liu**, Edward Lee, Mu Zhou, Aidong Lu, Shaoting Zhang*. Feature-enhanced graph networks for genetic mutational prediction using histopathological images in colon cancer[C]. International Conference on Medical Image Computing and Computer Assisted Intervention (***MICCAI***), 2020, (pp. 294-304). `https://doi.org/10.1007/978-3-030-59713-9_29`

30. Qingzhu Yang, **Qiao Liu**, Hairong Lv*. A decentralized system for medical data management via blockchain [J]. ***Journal of Internet Technology***, 2020, 21(5): 1335-1345. [link]

31. Junfeng Liu, **Qiao Liu**, Qingzhu Yang*. Mstree: a multispecies coalescent approach for estimating ancestral population size and divergence time during speciation with gene flow [J]. ***Genome Biology and Evolution***, 2020, 12(5): 715-719. `https://doi.org/10.1093/gbe/evaa087`

32. Chencheng Xu, **Qiao Liu**, Jianyu Zhou, Minzhu Xie, Jianxing Feng, Tao Jiang*. Quantifying functional impacts of regulatory variants with multi-task Bayesian neural network[J]. ***Bioinformatics***, 2020, 36(5): 1397-1404. `https://doi.org/10.1093/bioinformatics/btz767`

33. Chencheng Xu, **Qiao Liu**, Minlie Huang, Tao Jiang*. Reinforced molecular optimization with neighborhood-controlled grammars[C]. Conference on Neural Information Processing Systems (***NeurIPS***), 2020, 33. `https://doi.org/10.48550/arXiv.2011.07225`

34. **Qiao Liu**, Hairong Lv, Rui Jiang*. hicGAN infers super resolution Hi-C data with generative adversarial networks. ***Bioinformatics***, 2019, 35(14): i99-i107. Also in *Proceedings of the 27th Intelligent Systems for Molecular Biologythe 18th European Conference on Computational Biology* (***ISMB/ECCB***), 2019. `https://doi.org/10.1093/bioinformatics/btz317`

35. Qijin Yin, Mengmeng Wu, **Qiao Liu**, Rui Jiang*. DeepHistone: a deep learning approach to predicting histone modifications[J]. ***BMC Genomics***, 2019,20(2):193. `https://doi.org/10.1186/s12864-019-5489-4`

36. Shaoming Song, Hongfei Cui, Shengquan Chen, **Qiao Liu**, Rui Jiang*. EpiFIT: Functional interpretation of transcription factors based on combination of sequence and epigenetic information[J]. ***Quantitative Biology***, 2019, 1-11. `https://doi.org/10.1007/s40484-019-0175-8`

37. **Qiao Liu**, Fei Xia, Qijin Yin, Rui Jiang*. Chromatin accessibility prediction via a hybrid deep convolutional neural network[J]. ***Bioinformatics***, 2017, 34(5): 732-738. `https://doi.org/10.1093/bioinformatics/btx679`

38. **Qiao Liu**, Mingxin Gan, Rui Jiang*. A sequence-based method to predict the impact of regulatory variants using random forest[J]. ***BMC Systems Biology***, 2017, 11(2): 7. Also in *Proceedings of the 15th Asia Pacific Bioinformatics Conference* (***APBC***), 2017. `https://doi.org/10.1186/s12918-017-0389-1`

39. Bai Li, Mu Lin, **Qiao Liu**, Ya Li*, Changjun Zhou*. Protein folding optimization based on 3D off-lattice model via an improved artificial bee colony algorithm[J]. ***Journal of Molecular Modeling***, 2015, 21(10): 261. `https://doi.org/10.1007/s00894-015-2806-y`

Honors and Awards

NSF Travel Award — Jun 2026
ICSA Applied Statistics Symposium

Biostatistics Development Project Award — Feb 2026
Yale Pepper Center

Pathway to Independence Award (K99/R00) Aug 2024
National Institutes of Health (NIH), NHGRI

ECCB Fellowship Aug 2020
International Society for Computational Biology (ISCB)

ISMB Travel Award Apr 2019
International Society for Computational Biology (ISCB)

National Scholarship Oct 2018
Ministry of Education of China

Microsoft Young Fellowship Jun 2015
Microsoft Research Asia

Talks and Presentations

Invited Seminar Talks and Short Courses

NSF IMSI Workshop, University of Chicago, IL Oct 2026
AI-powered Bayesian generative modeling approaches for statistical inference

PharmaDS, Edison, NJ Mar 2026
CGM-AI: Causal Generalist Medical AI (1-day course)
Jointly with Dr. Hongtu Zhu

BIS-BIDS Seminar, Yale University, CT Dec 2025
Leveraging genomic LLMs and causal inference to elucidate complex disease mechanisms

DahShu Data Science Symposium, University of Connecticut, CT Oct 2025
CGM-AI: Causal Generalist Medical AI (1-day course)
Jointly with Dr. Hongtu Zhu

Duke Boot Camp for AI, Duke University, NC May 2025
CGM-AI: Causal Generalist Medical AI (1-day course)
Jointly with Dr. Hongtu Zhu, Dr. Haoxiu Yao, and Dr. Xin Wang

USC QCB Seminar, University of Southern California, CA Nov 2024
A Flexible Generative AI Framework for High-dimensional Data Analysis

Tsinghua Statistics Seminar, Tsinghua University (remote) Apr 2023
CausalEGM: A general causal inference framework by encoding generative modeling

CVI Early Career Research Roundtable, Stanford University, CA Jun 2022
The application of deep generative models in biology

1^{st} place in NeurIPS Multimodal Data Integration (remote) Dec 2021
Single cell multi-modal integrative analysis

CEGS seminar, Stanford University, CA Mar 2020
Simultaneous deep generative modelling and clustering of single-cell genomic data

Invited Conference Presentations

JSM, Boston, MA Aug 2026
AI-powered Bayesian generative modeling approaches for statistical inference

ICSA, Arlington, VA Jun 2026
A multi-modal diffusion model with dual-cross-attention for multi-omics data generation and translation

NESS, Stamford, CT May 2026
Bayesian Generative Modeling Approach for Statistical Inference

ASA-CT Mini-Conference, Stamford, CT May 2026
Bayesian Generative Modeling Approach for Statistical Inference

ENAR, Indianapolis Mar 2026
An AI-powered Bayesian generative modeling approach for statistical inference in observational studies

JSM, Nashville, TN Aug 2025
An AI-powered Bayesian generative modeling approach for causal inference in observational studies

STATGEN, University of Minnesota-Twin Cities, MN May 2025
Genomic large language models and the applications in Alzheimer's disease

MCBIOS, University of Utah, UT Mar 2025
Multi-modal diffusion model with dual-cross-attention for multi-omics data generation and translation

ICIBM, Houston, TX Oct 2024
EpiGePT: a pretrained transformer-based language model for context-specific human epigenomics

JSM, Washington DC Aug 2022
scTrace: Time series single-cell trajectory inference using deep generative models

NIH CEGS Annual Meeting, Duke University, NC Aug 2022

Analyzing spatially resolved transcriptomics data with deep generative models

ECCB, remote talk — Sep 2020
DeepCDR: A hybrid graph convolutional network for predicting cancer drug response

ISMB, Basel, Switzerland — Jul 2019
hicGAN infers super resolution Hi-C data with generative adversarial networks

APBC, Shenzhen, China — Jan 2017
A sequence-based method to predict the impact of regulatory variants using random forest

Mentoring

Postdoctoral Associate
Haochen Wang — 2025 -

Ph.D. Student Advisees
Zijing Gao (joint with Rui Jiang) — 2024 -
Erpai Luo (joint with Xuegong Zhang) — 2024 -
Xuejian Cui (joint with Rui Jiang) — 2024 - 2026

Ph.D. Dissertation Committee
Erica Stutz (Chair: Andrew Loza) — 2026 -

Master Student Thesis Advisees
Jingyi Guo (joint with Hongyu Zhao) — 2025 - 2026

Grants

YSPH Transformation Grant — Jul 2026 - Jun 2027
Liu (PI) — $80,000 (direct cost)
Trustworthy causal AI system for identifying disease-relevant variant effects through regulatory genomics

Biostatistics Development Project Award — Jul 2026 - Jun 2027
Liu (PI) — $58,625
Unraveling causal pathways in age-related cognitive decline with generative AI and causal inference

K99/R00HG013661 — Sep 2024 - Aug 2028
Liu (PI) — $878,076
Bridging the gap between genetic variants and radiomic phenotypes via genomic large language models

Teaching

STATS 371: Applied Bayesian Statistics — Spring 2025
Department of Statistics, Stanford University
Guest Lecturer

STATS 319: Literature of Statistics Spring 2024
Department of Statistics, Stanford University
Teaching Assistant

Introduction to Generative Models Summer 2023
ICME, Stanford University
Teaching Assistant

Intelligent Algorithms and Systems Fall 2019
Fundamental Industry Training Center, Tsinghua University
Teaching Assistant

Introduction to Artificial Intelligence Fall 2017
Department of Automation, Tsinghua University
Teaching Assistant

Service

Conference Session Chair
STATGEN, University of Minnesota, MN May 2025

Journal Reviewer
Annals of Applied Statistics
Bioinformatics
Biostatistics
Briefings in Bioinformatics
Bioinformatics Advances
BMC Bioinformatics
BMC Genomics
Complex & Intelligent Systems
Computational Statistics
Drug Discovery Today
Engineering Applications of Artificial Intelligence
Genome Biology
Genome Research
Genomics, Proteomics & Bioinformatics
IEEE Transactions on Medical Imaging
iScience
Journal of the American Statistical Association
Journal of Causal Inference
Journal of Computational and Graphical Statistics
Nature Methods
Nature Machine Intelligence
Nature Communications
NAR Genomics and Bioinformatics
Proceedings of the National Academy of Sciences

Patterns
PLOS Computational Biology
Statistics in Medicine
Transactions on Machine Learning Research

Conference Reviewer (PC Member)
Intelligent Systems for Molecular Biology (ISMB) 2026
Conference on Neural Information Processing Systems (NeurIPS) 2026
International Conference on Learning Representations (ICLR) 2025-26
Association for the Advancement of Artificial Intelligence (AAAI) 2023-25

PROFESSIONAL MEMBERSHIP

American Statistical Association (ASA)
International Society for Computational Biology (ISCB)
Eastern North American Region of the International Biometric Society (ENAR)
International Chinese Statistical Association (ICSA)